

Reflection Journal — Module 2 Lab Exercise Jonathon Wiltturner | ITAI 1371

Going into this lab, I expected the hardest part to be writing the code. It turned out that wasn't true. The code ran fine. What actually gave me trouble was understanding the graphs and the output once they showed up on the screen. Looking at a scatter plot or a block of statistics is different from writing the code that made them, and at first I wasn't sure what I was actually looking at. That confusion started to clear up once I slowed down and really looked at what the scatter plot was showing instead of just glancing at it. Once I understood that the plot was showing me which species of flower were easy to tell apart and which ones overlapped, the rest of the lab made a lot more sense. Before that moment, the graph was just a picture with dots on it. After that moment, it was information I could actually read. That was the real turning point for me in this assignment, not writing the code, but learning how to interpret what the code produced. Looking back at the specific observations I wrote in the notebook, I noticed Setosa formed its own clear group with almost no overlap with the other two species. Versicolor and Virginica overlapped more in the middle of the chart. Once I understood that overlap, I understood why sepal measurements alone wouldn't always be enough to tell those two species apart, which is something I wouldn't have picked up on just from looking at the raw numbers in a table. I also saw how pandas and NumPy work together during this lab. Pandas kept the data organized, keeping each measurement connected to the correct species name. NumPy handled the actual math, like calculating the mean and standard deviation. Before this lab those felt like two separate things I had to memorize. Now I understand pandas is more like the spreadsheet, and NumPy is the calculator working underneath it. The habit I'm actually taking with me from this lab is looking at the data before building anything. It would have been easy to skip straight to the statistics or jump into building something without stopping to visualize what the data actually looks like first. But seeing the scatter plot before anything else changed how I understood the rest of the lab. I plan to carry that habit forward, look at the data first, understand what's really there, before trying to build on top of it. This lab also left me with questions I didn't have going in. I'm curious how this same kind of visualization would hold up with a much larger, messier, real world dataset instead of a small clean sample like Iris. I'm also curious whether using petal length and width instead of sepal measurements would separate Versicolor and Virginica more clearly, since the overlap I saw made me want to test a different pair of features and see if the picture changes. This lab was small, but the biggest thing I'm taking from it is that reading and understanding output is its own skill, separate from writing the code. I can't just run something and move on. I have to actually look at what it produced and understand what it's telling me before I can call it done.

Reflective Learning Journal: Module 2 Lab Exercise

Course: ITAI-1371 Machine Learning

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1. Cognitive Breakthroughs & Conceptual Connections

Working through this lab provided a clear bridge between abstract numerical data and practical machine learning concepts. My biggest breakthrough occurred while working with Matplotlib scatter plots. Seeing feature values mapped visually transformed raw numbers into distinct geometric clusters. Watching the Setosa species isolate itself neatly from Versicolor and Virginica illustrated why feature selection is fundamental to classification

algorithms.

Moving from manual spreadsheet operations to Python's data stack demonstrated the power of programmatic data analysis. Using Pandas DataFrames made indexing and grouping straightforward, while NumPy vectorization eliminated the need for explicit loops, drastically improving computational clarity and efficiency.

2. Technical Challenges & Debugging Strategies

During the assessment tasks, working with array dimensions and statistical aggregations required careful attention. When calculating mean and standard deviation across specific DataFrame columns, I realized the importance of verifying data types and handling missing values before passing inputs into NumPy operations.

Understanding warning messages and output formatting also reinforced best practices in reproducible research. Suppressing non-critical warnings ensured clean output presentation, which is essential for professional documentation.

3. Evolution of Technical Thinking

Before this lab, I viewed data preprocessing as a tedious precursor to machine learning. However, visualizing feature distributions highlighted that model accuracy relies heavily on clean, well-structured inputs. Quality data preparation directly influences a model's ability to learn decision boundaries effectively.